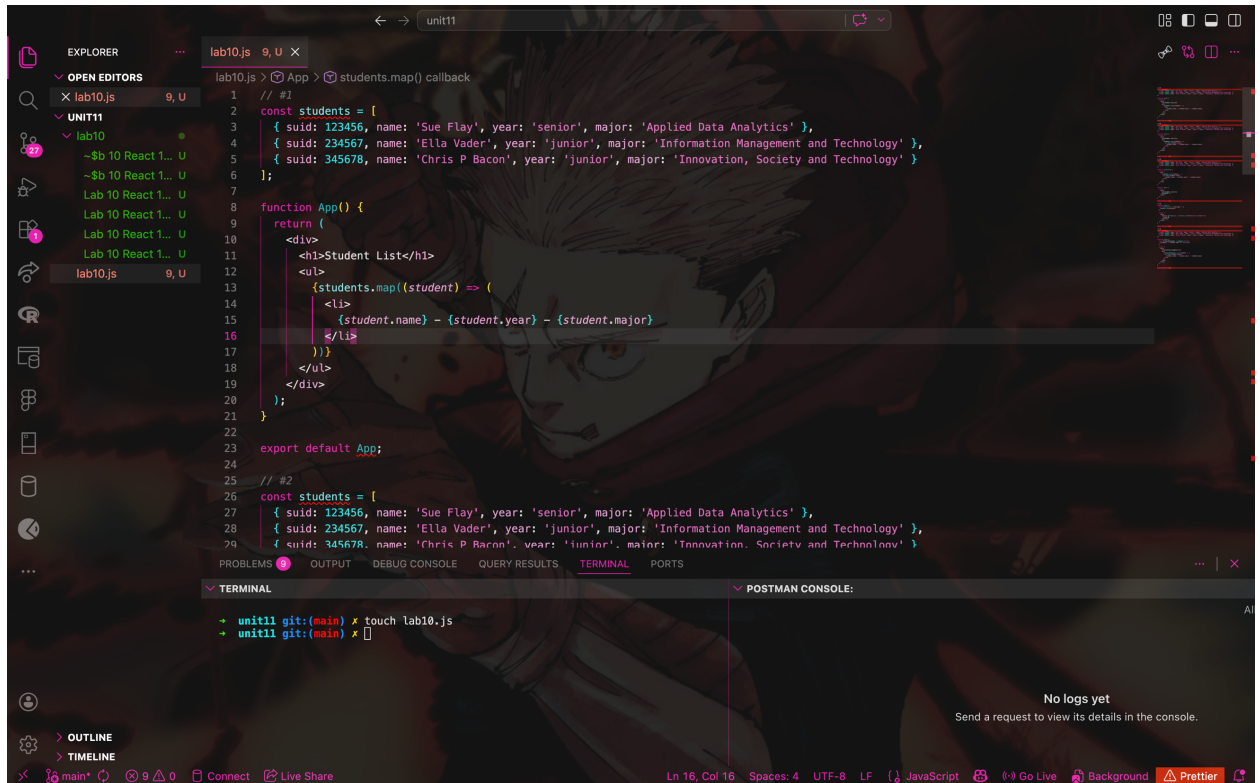


IST 363

LAB 10

REACT 1

1. paste screenshot here



```
1 // #1
2 const students = [
3   { suid: 123456, name: 'Sue Flay', year: 'senior', major: 'Applied Data Analytics' },
4   { suid: 234567, name: 'Ella Vader', year: 'junior', major: 'Information Management and Technology' },
5   { suid: 345678, name: 'Chris P Bacon', year: 'junior', major: 'Innovation, Society and Technology' }
6 ];
7
8 function App() {
9   return (
10     <div>
11       <h1>Student List</h1>
12       <ul>
13         {students.map((student) => (
14           <li>
15             {student.name} - {student.year} - {student.major}
16           </li>
17         ))}
18       </ul>
19     </div>
20   );
21 }
22
23 export default App;
24
25 // #2
26 const students = [
27   { suid: 123456, name: 'Sue Flay', year: 'senior', major: 'Applied Data Analytics' },
28   { suid: 234567, name: 'Ella Vader', year: 'junior', major: 'Information Management and Technology' },
29   { suid: 345678, name: 'Chris P Bacon', year: 'junior', major: 'Innovation, Society and Technology' }
30 ];
```

terminal

```
+ unit11 git:(main) x touch lab10.js
+ unit11 git:(main) x
```

2. paste screenshot here

The screenshot shows a VS Code editor window with the file `lab10.js` open. The code defines a `students` array and a `App` function. The `App` function uses `students.map` to render a list of students. The terminal shows the command `unit11 git:(main) x touch lab10.js`.

```
lab10.js > App > students.map() callback
21  function App() {
22  }
23  export default App;
24
25  // #2
26  const students = [
27    { suid: 123456, name: 'Sue Flay', year: 'senior', major: 'Applied Data Analytics' },
28    { suid: 234567, name: 'Ella Vader', year: 'junior', major: 'Information Management and Technology' },
29    { suid: 345678, name: 'Chris P Bacon', year: 'junior', major: 'Innovation, Society and Technology' }
30  ];
31
32  function App() {
33    return (
34      <div>
35        <h1>Student List</h1>
36        <ul>
37          {students.map((student) => (
38            <li key={student.suid}>
39              {student.name} - {student.year} - {student.major}
40            </li>
41          ))}
42        </ul>
43      </div>
44    );
45  }
46
47  export default App;
48
```

Terminal:

```
+ unit11 git:(main) x touch lab10.js
+ unit11 git:(main) x
```

3. paste screenshot here

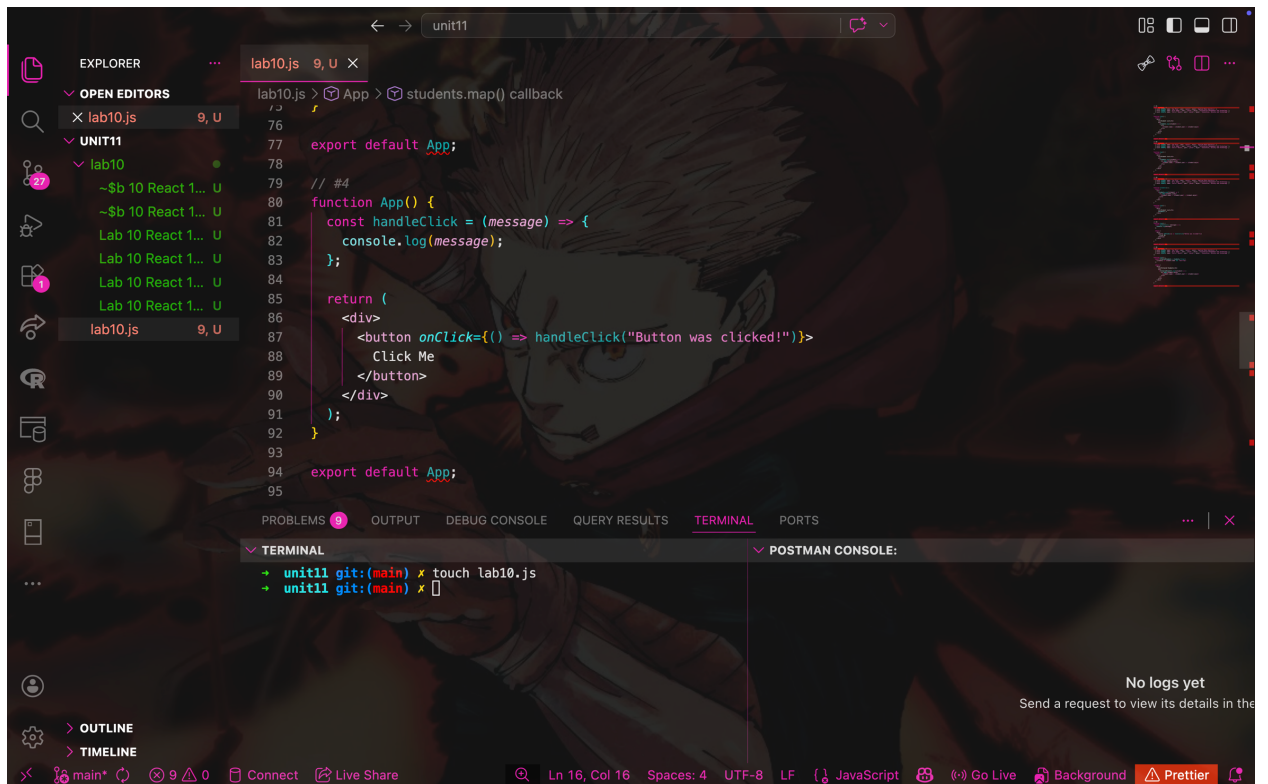
The screenshot shows a VS Code editor window with the file `lab10.js` open. The code defines a `Students` function and a `App` function. The `App` function uses `Students` to render a list of students. The terminal shows the command `unit11 git:(main) x touch lab10.js`.

```
lab10.js > App > students.map() callback
50  // #2
51  const students = [
52    { suid: 123456, name: 'Sue Flay', year: 'senior', major: 'Applied Data Analytics' },
53    { suid: 234567, name: 'Ella Vader', year: 'junior', major: 'Information Management and Technology' },
54    { suid: 345678, name: 'Chris P Bacon', year: 'junior', major: 'Innovation, Society and Technology' }
55  ];
56
57  function Students() {
58    return (
59      <ul>
60        {students.map((student) => (
61          <li key={student.suid}>
62            {student.name} - {student.year} - {student.major}
63          </li>
64        ))}
65      </ul>
66    );
67  }
68
69  function App() {
70    return (
71      <div>
72        <h1>Student List</h1>
73        <Students />
74      </div>
75    );
76  }
77
78  export default App;
79
```

Terminal:

```
+ unit11 git:(main) x touch lab10.js
+ unit11 git:(main) x
```

4. paste screenshot here



5. paste screenshot here

```
lab10.js 9, U X
lab10.js > App > students.map() callback
95 // #5
96 const students = [
97   { suid: 123456, name: 'Sue Flay', year: 'senior', major: 'Applied Data Analytics' },
98   { suid: 234567, name: 'Ella Vader', year: 'junior', major: 'Information Management and Technology' },
99   { suid: 345678, name: 'Chris P Bacon', year: 'junior', major: 'Innovation, Society and Technology' }
100 ];
101
102 function App() {
103   const filteredStudents = students.filter(
104     (student) => student.name === "Sue Flay"
105   );
106
107   return (
108     <div>
109       <h1>Filtered Students</h1>
110       <ul>
111         {filteredStudents.map((student) => (
112           <li key={student.suid}>
113             {student.name} - {student.year} - {student.major}
114           </li>
115         ))}
116       </ul>
117     </div>
118   );
119 }
120
121 export default App;
```

unit11 git:(main) x touch lab10.js
unit11 git:(main) x

No logs yet
Send a request to view its details in the console.

6. write brief answers to the questions below

a.

React components are reusable pieces of UI that help organize code and make apps easier to manage.

b.

They must start with uppercase so React knows it is a component and not a regular HTML tag.

c.

JSX is a syntax that looks like HTML but lets you write JavaScript inside it.

d.

We use className because class is a reserved word in JavaScript.

e.

You use curly braces `{}` to insert JavaScript inside JSX.

f.

It makes code cleaner, easier to read, and easier to reuse.

g.

When a component gets too big or does too many things.

h.

Yes, you can reuse components by calling them like multiple times.

i.

React handles inputs using state and events like `onChange` to track user input.

